

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS

FRIDAY, 30 SEPTEMBER 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **CHOOSE ANY ONE (1)** question in **Section B**.
Each question carries 25 marks each.

Note: Should you answer more than ONE (1) question in section B, marks will only be awarded to the **FIRST** question attempted.

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Section A (Answer ALL Questions)**

- Q1. (a) Table 1.0 shows the sequences of activities in the ABC project that focuses on the development of an information system.

Activity	Expected Time	Successor List
1. Programming & Testing	10	3
2. Requirement Collection	6	4
3. Installation	4	-
4. Requirement Analysis	3	1,5
5. Prototype	5	1

Table 1.0: Sequence of activities in ABC project.

- (i) Draw the network diagram for the activities listed in Table 1.0. (5 marks)

***Note: Activity 3: Installation is the final successor activity in your network diagram.**

- (ii) Calculate the Early Start (ES) and Late Start (LS) for activities 2, 3, and 5 listed in Table 1.0. (6 marks)

- (b) You are the owner of a tour agency, known as Grand Tourist Agency. You are planning to develop an interactive travel planning mobile application for your potential customers that will help to plan their journeys. Since you do not have expertise and sufficient resources in your company to develop the application, therefore you are planning to outsource the software development. Your initial requirement is to develop a custom application that contains login function, interactive travel planning to the desired location function, itinerary and quotation generation function, and tour booking and payment function. In your opinion, identify and indicate the most appropriate source of software that you will choose to outsource the interactive travel planning mobile application in accordance to your initial requirements. Justify your answers. (8 marks)

- (c) The process of requirements determination is conducted by the system analyst to gather information about an information system's project. Identify and indicate **THREE (3)** traditional methods for requirements determination. (6 marks)

[Total : 25 marks]

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS

- Q2. (a) During the development of an information system, test cases are usually developed by the system analysts and not the programmers. In your opinion, explain why this process is important for an efficient system development. (4 marks)
- (b) Training and support are practiced separately or can be combined as one process. According to the above statement, answer the following questions:
- (i) Suggest **THREE (3)** types of training methods for system users. (6 marks)
- (ii) Describe a Help Desk support provided to the system users. (6 marks)
- (c) Identify and indicate **ONE (1)** information system that you had designed in your assignment. As a developer of the system, explain how user demographic and system characteristic factors shall influence the users to use the system. (9 marks)
- [Total : 25 marks]
- Q3. (a) System maintenance is to repair or improve a system's functionalities. There are four types of system maintenances.
- (i) Discuss the differences between preventive and adaptive system maintenances. (8 marks)
- (ii) Imagine that you are a system maintenance personnel of XYZ organization. You had performed and installed a standalone system in your customer's organization, one year ago. In addition, as requested by the customer, you have been performing maintenance service to the system, periodically for every once in three months. However, recently your customer raised a concern that the response time of the system to users' requests is slow. The customer has requested you to improve the system's performance for fast response. In your opinion, according to your customer's request, will you perform corrective or perfective maintenance on the system? Justify your answers. (8 marks)
- (b) In a research study, it was found that 80% budget of an organization is used to maintain an information system.
- (i) Suggest **THREE (3)** factors that influence the increasing cost of system maintenance in an organization. (3 marks)

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Q3. (b) (Continued)**

- (ii) An information system maintenance request, usually, is submitted by the system users to the project manager. Later the project manager shall forward the request to the system priority board. Explain the process of system maintenance request after the request is forwarded to the system priority board. (6 marks)

[Total : 25 marks]

Section B (Choose any ONE question)

- Q4. (a) Swarz company owns an online car rental system to automate the rental car booking process to its customers. The customer has to login to the system to perform the booking. Upon successful login, the customer has to enter his personal particulars, driving license details, type of car model to be rented, number of days and time for rental. These details shall be stored as a record into the system database. The admin staff at the Swarz company will then verify these details for approval. Upon successful approval, notification will be sent to the customer to perform deposit payment via the system. Otherwise, the customer will be notified that the booking is unsuccessful with valid justification(s). For successful booking approval, once the customer has deposited the payment, a receipt is issued to the customer's email from the system. This receipt will be used by the customer during the rental car collection. Moreover, the manager at the Swarz company is able to view the daily rental cars booking report via the system.

According to the above scenario, prepare the items listed below to form a data flow diagram (DFD).

- (i) Determine **TWO (2)** sources/sinks and **ONE (1)** data store for the given scenario. (3 marks)
- (ii) Determine **FIVE (5)** processes for the given scenario. (5 marks)
- (iii) Draw a data flow diagram (DFD) level 0 using the listed sources/sinks and data store from Q4(a)(i) and the processes identified in Q4(a)(ii) in correct sequence. (9 marks)
- (b) Report is a passive business document that contains only predefined data. Determine **FOUR (4)** common types of reports. (8 marks)

[Total : 25 marks]

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS

- Q5. (a) ARNAX is a well-established software firm that mainly focuses on software development. The firm holds a policy that for each software development project, at least a minimum of two project managers will be assigned for managing the project. While each manager manages only one project for development. This is to increase the efficiency of the workflow and for speedy decision derivation. Each project has its own project id, name, type, and customer details. Further, each project manager is the internal stakeholder who holds a managerial position at the firm, has manager id, and is from the Information System (IS) department. In addition, each project manager is responsible to select and lead all the team members to work for the project he/she manages. Each team member is a permanent staff who has staff id and is working in various departments at the firm. Their designation is either a system developer, system engineer, system analyst or system designer at the firm. Moreover, each team member is not necessarily led by one project manager. They may be led by a maximum of two managers who manage the same project.

According to the above scenario, prepare the items listed below to perform normalization using third normal form (3NF).

- (i) Determine **THREE (3)** relations for the given scenario. (3 marks)
 - (ii) Determine **First Normal Form (1NF)** for the relations identified in Q5(a)(i). (4 marks)
 - (iii) Determine **Second Normal Form (2NF)** for the 1NF relations identified in Q5(a)(ii). (4 marks)
 - (iv) Determine **Third Normal Form (3NF)** for the 2NF relations identified in Q5(a)(iii). (4 marks)
 - (v) Determine **the associative entity(s)** for the 3NF relations identified in Q5(a)(iv). (4 marks)
- (b) Hardware devices are commonly used for interaction with the interface of a system. The selection of these hardware devices is done during the logical design. Determine **THREE (3)** hardware devices used for the interaction.

(6 marks)

[Total : 25 marks]

